

POST-HOC EXPLANATION OF BLACK-BOX MODELS

Explainable AI (XAI) Project Report & Faculty Presentation Guide

1. Executive Summary & Core Objective

Modern Machine Learning models (like XGBoost and Deep Neural Networks) yield exceptional accuracy but function as **Black-Box Models** because their inner mathematical logic is opaque to human interpretation. This project implements a comprehensive **Post-hoc Explainable AI (XAI) Suite** and **Interactive Web Dashboard** to interpret black-box predictions without altering underlying model structures.

2. Trained Classification Models Performance

Model Architecture	Model Type	Accuracy	F1-Score	ROC-AUC
XGBoost Classifier	Black-Box (Ensemble)	88.33%	0.9067	0.9674
Multi-Layer Perceptron	Black-Box (Neural Net)	90.00%	0.9221	0.9641
Random Forest	Black-Box (Ensemble)	84.17%	0.8758	0.9384
Logistic Regression	White-Box (Baseline)	87.50%	0.9032	0.9419

3. Implemented Post-hoc Interpretability Techniques

- A. SHAP (SHapley Additive exPlanations):** Game-theoretic coalitional attribution calculating exact Shapley values. Satisfies Additivity Theorem: $\text{Base Value} + \text{sum}(\text{SHAP}) = \text{Prediction}$.
- B. LIME (Local Interpretable Model-agnostic Explanations):** Generates local sparse linear surrogate models around perturbed instance neighborhoods. Evaluated using local R^2 fidelity scores ($R^2 = 0.60 - 0.91$).
- C. Partial Dependence Plots (PDP) & ICE:** Visualizes marginal response curves illustrating feature interaction trends on predicted disease probabilities.
- D. Counterfactual Explanations:** Optimization-based minimal actionable feature modifications required to flip high-risk predictions to low-risk.

4. Simple Hinglish Explanation Guide for Faculty Presentation

Q1. Project ka main purpose kya hai?

Answer: Sir/Ma'am, jab hum complex ML models (jaise XGBoost ya Neural Networks) use karte hain, toh wo 'Black-Box' hote hain — yaani wo high accuracy toh dete hain par bata nahi sakte ki decision KYUN liya. Our project applies **Post-hoc XAI techniques** to explain black-box decisions after prediction without modifying the trained model.

Q2. Konsi 4 main post-hoc techniques implement ki hain?

Answer:

- SHAP:** Game theory based attributions jo exact feature impact point dikhati hai.
- LIME:** Prediction ke aaspas local linear model fit karke local feature weights + R^2 fidelity score batata hai.
- PDP / ICE:** Single feature change hone par overall risk curve dikhaata hai.
- Counterfactuals:** Actionable recommendations deta hai (e.g., 'If ST depression drops from 2.8 to 0.8mm, disease

risk drops to 22%').

Q3. Implementation & GitHub details?

Answer: Humne Flask REST API + HTML5/CSS3 glassmorphism UI + Plotly.js se live dashboard banaya hai. Entire project is pushed to GitHub: [**https://github.com/NehaMusale11/XAI-PROJECT-**](https://github.com/NehaMusale11/XAI-PROJECT-)